

Part-D

Project Work



Projects

Project 1: Create a document in Writer and perform the following actions:

1. Create the following table and format it.
2. Enter the days of your birthday month.
3. Make 11 copies of this table and enter the days of all the months of the current year.
4. Save the document with the title "Yearly Calendar".

MONTHLY CALENDAR

MONTH: _____ YEAR: _____

MONDAY

TUESDAY

WEDNESDAY

THURSDAY

FRIDAY

SATURDAY

SUNDAY

Project 2: Create the following invitation card in Writer and send it to your nearby schools using Mail Merge. You can change the colour and font size according to your requirements. Design the card by applying the background colour, borders, and inserting images.

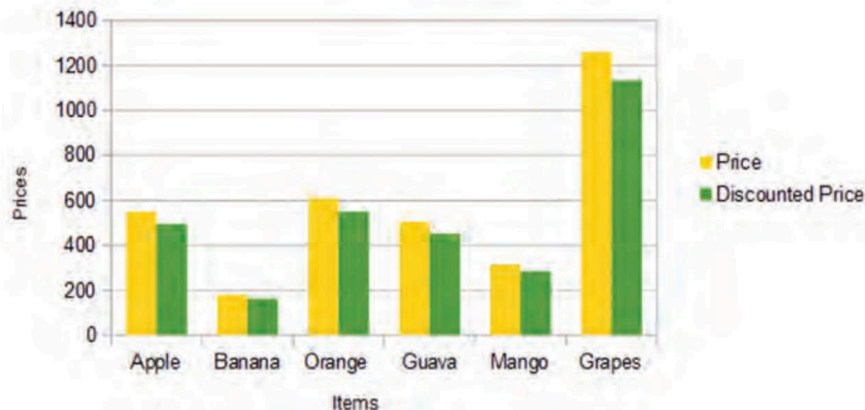


Project 3: Follow the given steps to calculate the discounted price for various fruit items.

1. Type the data for the various **Items**, **Units**, and **Price per unit** as given.
2. Click on the cell D6 and type the formula **=B6*C6** to calculate the price of the item Apple, and then press the **Enter** key.
3. Click on cell D6 again and drag the fill handle to copy the formula in all the cells upto D11.
4. Click on cell E6 and type the formula **=D6-(D6*\$D\$3)** to calculate the discounted price for the item Apple and press the **Enter** key.
5. Click on cell E6 again and drag the fill handle to copy the formula in all the cells upto E11.
6. Click on cell E14 and type the formula **=SUM(E6:E11)** to calculate the total discounted price.
7. Change the background colour of some of the cells as shown in the figure using the **Background Color** tool from the Formatting toolbar.
8. Wrap text in cell D14 by right-clicking the cell and selecting the **Format Cells** option. Click the **Alignment** tab in the **Format Cells** dialog box and choose the **Wrap text automatically** option. Click on **OK**.
9. Now, create a chart to visualise and analyse the data. Add the title, X axis, and Y axis chart elements to the chart.
10. Save the file with a suitable name and close the application.

	A	B	C	D	E
1			Total Price of Items		
2					
3			Discount	10.00%	
4					
5	Items	Units	Price per unit	Price	Discounted Price
6	Apple	10	55	550	495
7	Banana	15	12	180	162
8	Orange	18	34	612	550.8
9	Guava	12	42	504	453.6
10	Mango	9	35	315	283.5
11	Grapes	28	45	1260	1134
12					
13					
14				Total Discounted Price	3078.9

Total Price of Items



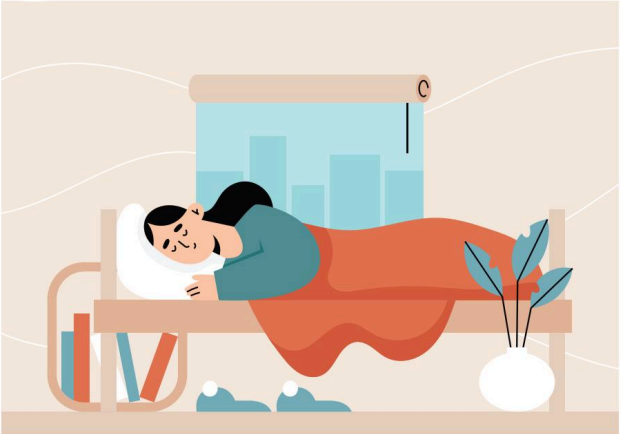
Project 4: Open the Calc spreadsheet and follow the given instructions.

1. Click on cell A1 and type "Daily Activities and Meal". Select the cells A1 to F1 and click on the **Merge Cells** option from the Formatting toolbar. Also, click the **Align Center Horizontally** option from the Formatting toolbar to centre align the text.
2. Click on cell A4 and enter the date 01/11/23. Drag the Fill handle upto cell A10. This will generate dates in the sequence automatically.
3. Similarly, click on the cell B4 and type Wednesday. Drag the Fill handle upto cell B10 to generate the days of the week in the sequence automatically.
4. Type the remaining data from the columns Weather, Activities, Meal Type, and Productivity Score.
5. Change the background colour of some of the cells as shown in the figure using the **Background Color** tool from the Formatting toolbar.
6. Click on cell D12 and type the formula **=MAX(F4:F10)** to find the maximum productivity score for the given days. Press the **Enter** key.
7. Similarly, click on cell D13 and type the formula **=MIN(F4:F10)** to find the minimum productivity score for the given days. Press the **Enter** key.
8. Click on cell D14 and type the formula **=AVERAGE(F4:F10)** to find the average productivity score for the given days. Press the **Enter** key.
9. Right-click the cell and select the **Format Cells** option. Click on the **Numbers** tab from the **Format Cells** dialog box. In the **Decimal places** option, choose 2 as the number of decimal places using the spin arrow.
10. Save the file with a suitable name and close the application.

	A	B	C	D	E	F
1	Daily Activities and Meal					
2						
3	Date	Day	Weather	Activities	Meal Type	Productivity Score
4	01/11/23	Wednesday	Sunny	Cycling	North Indian	7
5	02/11/23	Thursday	Cloudy	Walking	South Indian	8
6	03/11/23	Friday	Rainy	Watching Movie	Indo-Chinese	9
7	04/11/23	Saturday	Sunny	Outdoor games	Indo-Mexican	7
8	05/11/23	Sunday	Humid	Family Time	South Indian	6
9	06/11/23	Monday	Foggy	Rest Day	North Indian	8
10	07/11/23	Tuesday	Sunny	Work	North Indian	8
11						
12	Maximum Productivity Score			9		
13	Minimum Productivity Score			6		
14	Average Productivity Score			7.57		
15						

Project 5: Create a presentation on the topic “Tips for Good Health”.

1. The presentation should contain at least ten slides.
2. Every slide must have an image and a tip for good health.
3. Enter a related title on each slide.
4. Apply the same theme to all slides.
5. Some sample slides are given here:

Eat the Rainbow Consume a variety of colourful fruits and vegetables. 	Exercise Daily Make exercise a part of your daily routine. 
Balanced Diet Eating a variety of foods ensures your body gets all the necessary nutrients. 	Sleep Well Consume a variety of colourful fruits and vegetables. 

Project 6: Create a presentation on the topic “Famous Indian Scientists and Their Discoveries”.

1. Each slide should contain a picture of the scientist and information about one discovery.
2. Enter the name of the scientist as the title of the slide.
3. Apply custom animation to the objects of the slides.
4. Add the transition effect of your choice to each slide.

Sample Paper

Information Technology (SUBJECT CODE - 402)

Class IX (Session 2023-2024)

Max. Time: 2 Hours

Max. Marks: 50

General Instructions:

1. Please read the instructions carefully.
2. This Question Paper consists of 21 questions in two sections: Section A & Section B.
3. Section A has Objective type questions whereas Section B contains Subjective type questions.
4. Out of the given (5 + 16 =) 21 questions, a candidate has to answer (5 + 10 =) 15 questions in the allotted (maximum) time of 2 hours.
5. All questions of a particular section must be attempted in the correct order.
6. **SECTION A - OBJECTIVE TYPE QUESTIONS (24 MARKS):**
 - i. This section has 05 questions.
 - ii. Marks allotted are mentioned against each question/part.
 - iii. There is no negative marking.
 - iv. Do as per the instructions given.
7. **SECTION B - SUBJECTIVE TYPE QUESTIONS (26 MARKS):**
 - i. This section has 16 questions.
 - ii. A candidate has to do 10 questions.
 - iii. Do as per the instructions given.
 - iv. Marks allotted are mentioned against each question/part.

SECTION A: OBJECTIVE TYPE QUESTIONS

Q1. Answer any 4 out of the given 6 questions on Employability Skills.

(1 x 4 = 4 marks)

1. Which of the following element is not a part of non-verbal communication?

a. Gestures

☐

b. Eye contact

☐

c. Video calls

☐

d. Posture

☐

2. 'Hurray! We won the match.' Which type of sentence is this?

a. Assertive

☐

b. Exclamatory

☐

c. Imperative

☐

d. Interrogative

☐

3. _____ is one of the first and oldest ICT tools.

a. Radio

☐

b. Email

☐

c. Smartphones

☐

d. Tablets

☐

4. Which of the following factors can impact your self-confidence?

a. Physical

☐

b. Cultural

☐

- c. Social ☐ d. All of these ☐
5. Rajan's company buys products or goods from manufacturers at a wholesale rate and sells them to consumers at a retail price. Which type of business is this?
- a. Service ☐ b. Merchandising ☐
- c. Manufacturing ☐ d. Hybrid ☐
6. Which factor can cause an imbalance in the environment?
- a. Afforestation ☐ b. Conservation of resources ☐
- c. Global warming ☐ d. Proper use of water resources ☐

Q2. Answer any 5 out of the given 6 questions.

(1 x 5 = 5 marks)

1. _____ is one of the common mouse actions.
- a. Comfort ☐ b. Double Click ☐
- c. Speed ☐ d. Accuracy ☐
2. Which of the following is not a presentation software?
- a. Google Slides ☐ b. Microsoft PowerPoint ☐
- c. LibreOffice Impress ☐ d. OpenOffice Calc ☐
3. It consists of thumbnail images of the presentation's, slides arranged in the order that they will be displayed.
- a. Slides Pane ☐ b. Workspace ☐
- c. Side Bar ☐ d. Slide show ☐
4. In the formula, =\$D3+E5, which reference is absolute?
- a. Column E ☐ b. Column D ☐
- c. Row 5 ☐ d. Row 3 ☐
5. Which key combination moves the cell to A1 position in Calc?
- a. Ctrl + Home ☐ b. Ctrl + End ☐
- c. Shift + End ☐ d. Shift + Home ☐
6. The _____ allow you to navigate between different worksheets within a workbook.
- a. Navigation buttons ☐ b. Horizontal Scrollbar ☐
- c. Vertical Scrollbar ☐ d. Status Bar ☐

Q3. Answer any 5 out of the given 6 questions.

(1 x 5 = 5 marks)

1. Which of the following are global companies with operations in multiple countries?
- a. GICs ☐ b. ISPs ☐
- c. MNCs ☐ d. None of these ☐
2. What does the "Auto Recovery" feature in OpenOffice Writer do?
- a. Automatically corrects spelling errors ☐
- b. Automatically saves the document at regular intervals ☐
- c. Suggests synonyms for words ☐
- d. Embeds objects in the document ☐

3. You can create a table using the _____ shortcut key.
- | | | | |
|---------------|--------------------------|---------------|--------------------------|
| a. Ctrl + F9 | ☐ | b. Ctrl + F10 | ☐ |
| c. Ctrl + F11 | ☐ | d. Ctrl + F12 | ☐ |
4. Which of the following is the shortcut key to insert a new slide?
- | | | | |
|-------------|--------------------------|-------------|--------------------------|
| a. Ctrl + M | ☐ | b. Ctrl + N | ☐ |
| c. Ctrl + W | ☐ | d. Ctrl + O | ☐ |
5. _____ aligns the text evenly on both the left and right margins of a cell.
- | | | | |
|------------------------------|--------------------------|----------------|--------------------------|
| a. Align left | ☐ | b. Align right | ☐ |
| c. Align Center Horizontally | ☐ | d. Justified | ☐ |
6. _____ are the bars, slices, or other elements that show the data values.
- | | | | |
|---------------|--------------------------|----------------|--------------------------|
| a. Chart Wall | ☐ | b. Data series | ☐ |
| c. Legend | ☐ | d. Gridlines | ☐ |

Q4. Answer any 5 out of the given 6 questions.

(1 x 5 = 5 marks)

1. What is the full form of ECG Machine?
- | | | | |
|-------------------------------|--------------------------|-------------------------------|--------------------------|
| a. Electrocardiogram Machine | ☐ | b. Electrocardiogram Machine | ☐ |
| c. Electroncardiogram Machine | ☐ | d. Electroncardiogram Machine | ☐ |
2. _____ is a pointing device used to select a displayed menu item or draw pictures on a monitor.
- | | | | |
|--------------|--------------------------|-------------|--------------------------|
| a. Light pen | ☐ | b. Keyboard | ☐ |
| c. Joystick | ☐ | d. OMR | ☐ |
3. You can adjust _____ to control the amount of space between lines of text.
- | | | | |
|----------------------|--------------------------|-----------------|--------------------------|
| a. Alignment | ☐ | b. Indentation | ☐ |
| c. Paragraph spacing | ☐ | d. Line spacing | ☐ |
4. What is the purpose of Mail Merge?
- | | |
|---|--------------------------|
| a. To play games | ☐ |
| b. To create personalised documents for multiple recipients | ☐ |
| c. To edit images | ☐ |
| d. To browse the internet | ☐ |
5. The _____ function gives us the numeric count of the entered values of the selected cells.
- | | | | |
|------------|--------------------------|--------------|--------------------------|
| a. Sum() | ☐ | b. Average() | ☐ |
| c. Count() | ☐ | d. Max() | ☐ |
6. In OpenOffice Impress, how do you adjust the width of a column in a table?
- | | |
|---|--------------------------|
| a. Drag the column boundary | ☐ |
| b. Right-click and select 'Adjust Column Width' | ☐ |
| c. Press Ctrl + W | ☐ |
| d. Use the Format menu | ☐ |

Q5. Answer any 5 out of the given 6 questions.

(1 x 5 = 5 marks)

1. What is the purpose of the "Navigator" tool in OpenOffice Writer?
- | | |
|---------------------------|--------------------------|
| a. To browse the internet | ☐ |
|---------------------------|--------------------------|

- b. To navigate between different sections of a document ☐
- c. To edit images ☐
- d. To send emails ☐
2. In Writer, Find & Replace option is available in menu.
- a. File ☐ b. Edit ☐
- c. View ☐ d. Insert ☐
3. How can you create a new Slide Master in Impress?
- a. Click the Apply to All Slides button. ☐
- b. Click the Format menu and then select Slide Design. ☐
- c. Right-click a slide and then select Set Background Picture. ☐
- d. Use the Slide Transition icon on the sidebar. ☐
4. Which feature allows you to align the text?
- a. Text Organizer ☐ b. Text Alignment ☐
- c. Text Effects ☐ d. Text Animation ☐
5. allows you to print in a specific order when printing multiple copies of a document.
- a. Orientation ☐ b. Paper source ☐
- c. Collate ☐ d. Range and copies ☐
6. A cell or a range of cells included in a formula for performing calculations are known as
- a. Constants ☐ b. Functions ☐
- c. References ☐ d. Mathematical Operators ☐

SECTION B: SUBJECTIVE TYPE QUESTIONS

Answer any 3 out of the given 5 questions on Employability Skills.

(2 x 3 = 6 marks)

Answer each question in 20–30 words.

- Q6. How is self-management beneficial for us? Give any two reasons.
- Q7. List any two advantages of visual communication.
- Q8. Differentiate between RAM and ROM.
- Q9. What is a partnership business?
- Q10. Explain how entrepreneurship benefits society economically?

Answer any 4 out of the given 6 questions in 20–30 words each.

(2 x 4 = 8 marks)

- Q11. Explain the significance of using templates.
- Q12. What is a Slide pane?
- Q13. What are modifier keys?
- Q14. Define Standard toolbar.
- Q15. Differentiate between front office and back office outsourcing.
- Q16. Differentiate between merging and splitting cells.

Answer any 3 out of the given 5 questions in 50–80 words each.

(4 x 3 = 12 marks)

- Q17.** Megha wants to know how information technology (IT) has benefited the banking sector. Explain this by providing examples of IT applications in the banking sector.
- Q18.** Girdhari is not aware of basic mouse actions. Explain it to him by describing any four.
- Q19.** Shanu wants to know the purpose of the Auto Spell Check feature. Help her by explaining the concept.
- Q20.** Rohan wants to rename a slide. Help him by mentioning all the steps involved.
- Q21.** Radha wants to remove an animation effect from an object on the slide. Help her by writing the steps to remove animation effects.

For Visually Impaired Students

Q.21 Define the following:

- | | |
|-----------------------|-------------------------|
| a. Formatting | b. Scatter chart |
| c. Indentation | d. Document area |

Employability Skills

Unit 1: Communication Skills-I

Answer Key

- A.** 1. d 2. c 3. b 4. a
5. c 6. a 7. b 8. d
- B.** 1. Articles 2. symbols 3. subject 4. receiver 5. Visual perception

- C.** 1. False. A letter is a written, typed or printed message that is enclosed in an envelope and sent through a postal service.
2. False. A noisy or chaotic environment will be a barrier to a communication process.
3. True.
4. False. The article 'an' should be used before the noun 'hour'.
5. True.

- D.** 1. Encoding includes putting ideas or information in a format that can be transmitted confidentially to a receiver. This often involves selecting words, creating visuals, and using symbols, gestures, or body language to convey the intended message accurately.

Decoding is the process by which the receiver interprets the encoded message and extracts its meaning. This process involves understanding the words, symbols, or gestures used by the sender and translating them into thoughts or emotions.

2. Advantages of visual communication are:

Makes communication meaningful	Use of graphics, diagrams, charts or models makes communication effective and purposeful.
Easy comprehension	Support of visual aids makes information easy to understand.
Attention grabbing	Visual communication, when coupled with verbal communication, makes it more appealing and attention-grabbing.

3. Articles are of two types: Definite and Indefinite articles.

a. Definite Article (the)

We use 'the' when we are talking about a specific thing that the listener or reader already knows about or can easily identify.

Example: The manager called a staff member to carry the files. (*a particular manager*)

b. Indefinite Articles (a/an)

We use 'a' or 'an' when we are talking about something in a general or non-specific way.

Example: I have a meeting at 2 pm.

E. 1.

Basis	Verbal Communication	Non-Verbal Communication
Meaning	Involves using spoken or written words to convey messages and information.	Conveys meaning through facial expressions, gestures, body language and other non-verbal cues.
Use of Language	Requires the use of language, including grammar and vocabulary.	Does not use language directly but involves elements like tone, volume and pace.
Forms	Face-to-face conversations, phone calls, speeches and written documents.	Facial expressions, gestures, posture, eye contact and touch.
Medium	Utilises spoken words, written documents, phones and computers.	Uses body language, facial expressions, gestures and physical presence.
Example	Writing an email or giving a speech.	Nodding head or shaking hands.

2. Any three factors which affect perspectives in communication are:

- **Language:** The language one uses to express themselves and their proficiency in a language can influence communication. When one employs incorrect terminology or unfamiliar jargon, or fails to provide sufficient details, a language can act as a barrier to communicating what one wishes to convey.
- **Experiences:** The experiences of the past give shape to a person's communication habits, triggers and sensitivities. A traumatic experience from the past can stop the person from understanding or communicating clearly or without hesitation.
- **Prejudices:** Prejudices, or previously built notions about individuals, groups or a thing can affect how we communicate and express our views. If we have biases against a particular group, we may be less likely to listen to their perspectives or give them the benefit of the doubt.

3. A sentence is a group of words that combines to express a complete thought. A sentence conveys a complete thought or idea and often serves as a complete statement or question. A sentence always starts with a capital letter and ends with a full stop, question mark or exclamation mark.

A sentence contains a subject and a predicate. A **subject** in a sentence is the thing or person about whom we get information from the sentence. A **predicate** in a sentence is what is being said about the person or thing in the sentence.

Subject (Who/what the sentence is about)	Predicate (What is being said about the subject)
Rajesh	loves to play cricket.

F. Competency-based questions.

1. Good communication skills help us in:

- **Building relationships:** Effective communication helps in establishing and maintaining strong relationships, whether with friends, family or colleagues. It fosters trust and understanding.
- **Effective interaction:** Communication skills are important when we interact with others, whether in personal, professional or social contexts. We need to express our thoughts and ideas clearly for a meaningful exchange.
- **Problem-solving:** When faced with challenges, effective communication is crucial for finding solutions. Misunderstandings can escalate into larger problems. To resolve conflicts, individuals should express their concerns, understand others' views and find mutually agreeable solutions to problems.
- **Sharing knowledge:** Through communication, we can pass knowledge from one generation to the next, ensuring that society continues to learn and grow.

- **Leadership and influence:** People who can communicate well generally become effective leaders, as they are able to convey their vision clearly, inspire others and influence decisions within an organisation.
2. A well-structured paragraph typically consists of three essential elements:
- **A topic sentence:** The topic sentence is like the headline of the paragraph. The sentence introduces the main idea or point that the paragraph will discuss. A topic sentence is a crucial element that guides the reader's understanding of what is to come.
 - **Supporting details:** Supporting details are sentences or examples that provide evidence or an explanation for the topic sentence. They add depth and context to your main idea.
 - **A concluding sentence:** The concluding sentence summarises the main point of the paragraph and often provides a smooth transition to the next paragraph.

Unit 2: Self-management Skills-I

Answer Key

- A.** 1. b 2. a 3. d 4. b 5. c
- B.** 1. Self-management 2. cultural 3. achievable 4. society 5. Time management
- C.** 1. True.
2. True.
3. False. One should try to appreciate the little joys in life.
4. False. Self-management is about navigating through all uncertainties or challenges.
5. False. Self-confidence is a multi-faceted concept that is influenced by physical, social, and cultural factors.
- D.** 1. Self-management is the capability of an individual to exercise control over one's feelings, ideas, thoughts, emotions, and behaviour to achieve the desired goals in both personal and professional settings. It helps an individual make effective decisions and maintain their well-being. It includes working towards the pre-determined goals and motivating oneself. Those who are better at managing themselves are able to set their priorities, manage their time, control their impulses, and take responsibility for their actions. They are more productive and content in life.
2. One should stay clean, hygienic, and smart as:
- Every individual should maintain good personal hygiene, cleanliness, and a well-groomed appearance. It will boost their confidence and help them communicate well with others.
 - Everyone must maintain a daily routine to take care of their hygiene and also try to dress appropriately according to the context of the situation.
 - Regular physical activity and having a balanced diet will contribute to overall well-being and appearance.
3. A self-confident person believes that they are capable of achieving their goals and will be an asset to any organisation. They typically possess these three qualities:
- Trust in their own capability
 - Readiness to accept new challenges
 - Willingness to take risks
- E.** 1. The way we perceive ourselves and the world is significantly influenced by our earliest interactions with the society around us. In this way, society acts like a mirror, reflecting and shaping our self-perception. Therefore, society plays a vital role in moulding and nurturing our self-confidence.
- Our sense of self can be impacted by the opinions of others, such as our parents, teachers, classmates, and friends. This can lead to an increase or decrease in self-confidence over time. If our parents support us, we will be able to excel in whatever we do, as their positive feedback will boost our confidence.
 - Also, external pressures, comparisons, discouraging comments, lack of attention, punishments, excessive criticism, and negative remarks can weaken an individual's self-belief and confidence. It can also break a person's confidence.

2. The various skills that can help manage the overall personality of a person include:

Problem Solving	It is to identify problems and challenges and, thereby, try to find practical solutions to solve them. It includes analysing issues and implementing strategies to resolve them.
Teamwork	It is to collaborate and work together as a team, and believe in the collaborative effort of different individuals who work towards the same goal.
Time Management	It is the ability to successfully complete tasks within specified deadlines and the ability of an individual to minimise distractions and unproductive activities.
Goal Setting	It requires one to plan clear and achievable goals that can be accomplished and provide motivation for personal or professional growth.
Decision Making	It is an individual's ability to make informed decisions by evaluating options and consequences and taking the most appropriate way forward.
Stress Management	It is the coping mechanism that enables an individual to handle work efficiently, even under pressures or difficulties.

3. Tips to build self-confidence in an individual are:

- **Get rid of negative thoughts:** Getting rid of negative thoughts is a very significant step in building one's self-confidence and maintaining it. A person has to change their mindset and shift their focus to more positive things in life.
- **Think positively:** Positive thinking involves focusing on the positives, understanding, and working towards improvement rather than focusing on the negatives. This mindset will help a person believe in their abilities, take on challenges, and achieve their goals.
- **Find happiness in small things:** One should try to appreciate the little joys in life, as it will cultivate a positive mindset and enhance their self-confidence. An individual can reflect on small positive things in their daily life, as it would boost their mood, help them appreciate present moments, and increase their self-awareness.
- **Chat with positive people:** Positive individuals provide encouragement, inspiration, and support, which will help other people feel more confident. They will help others believe in their abilities, reinforce their achievements, and create a supportive and uplifting atmosphere.

F. 1. The self-management skills that will surely help Tanuj are:

- to have clear objectives in mind and how he has to move towards his end goals.
 - to allocate his time wisely for tasks that are of priority to avoid unnecessary delays.
 - to reduce the stress of pending tasks as they will be dealt with according to priority.
 - to plan his schedule wisely and allot time slots to different activities in a day.
 - to maintain focus, not pay attention to distractions, and stay committed to his tasks and goals.
2. Kashika's confidence has been impacted by negative surroundings. Being surrounded by individuals with a consistently negative attitude was beginning to influence her mindset. Mocking others and pessimistic viewpoints have taken deeper roots into her thoughts and self-perception, impacting her self-confidence. Constant exposure to negativity can lead us to question our abilities and values.

Unit 3: ICT Skills-I

Answer Key

- A.** 1. a 2. d 3. a 4. b 5. a
- B.** 1. Luminance 2. Information and Communication Technology 3. Projector
4. Microphone 5. Application
- C.** 1. True.
2. True.
3. True.
4. False.
5. False.
- D.** 1. ICT has enabled online banking, making payments through mobile apps, and managing finances efficiently.
2. SD cards are small, removable storage devices that are commonly used in cameras and other portable devices. They are available in a wide range of sizes and capacities, making them a flexible storage option for users who need to store and transfer data on the go.
3. There are three types of ROM:
• PROM (Programmable ROM)
• EPROM (Erasable PROM)
• EEPROM (Electrically EPROM)
- E.** 1. A modem, short for modulator-demodulator, is a hardware device that allows a computer or other devices to connect to the internet. It modulates digital data into an analog signal that can be transmitted over telephone lines, cable lines, or fibre optics, and then demodulates the incoming analog signal back into digital data. There are several types of modems, including dial-up modems, DSL modems, cable modems, and fibre modems.
2. Following is the role of ICT in personal life:
• **Entertainment:** Television, radio, tablets, computer applications, and web browsers are useful for watching movies, listening to music and news, etc.
• **Money management:** Online banking apps have made the sending and receiving of money secure and convenient. It can be used for paying bills.
• **Travel:** ICT has enabled people to travel across the globe with the easy booking of train/bus/aeroplane tickets and hotels.
• **Learning and information:** Internet enables us to watch educational videos, find answers to questions, and learn new things online.
3. **System software:** System software controls the overall working of a computer. It manages all the input and output operations of the computer. For example, the operating system is a part of the system software that makes a computer run smoothly.
Application software: Application software facilitates fundamental computer operations. It performs specific tasks for users. This category includes word processors, spreadsheets, and a variety of other task-specific programs.

F. Competency-based questions.

1. Given below are the steps to start a computer:
 - Connect the power cable and peripherals.
 - Press the power button to start the computer.
 - Wait for the operating system to load.
2. Light Pen.

Unit 4: Entrepreneurial Skills-I

Answer Key

- A.** 1. d 2. c 3. b 4. d 5. b
- B.** 1. wholesale 2. Entrepreneurship 3. restaurant 4. influential 5. entrepreneur
- C.** 1. False. A sole proprietorship is a straightforward business structure where an individual single-handedly owns and manages the entire enterprise.
2. True.
3. False. Any new business initiative carries a considerable chance of failure.
4. True.
5. False. Entrepreneurs positively impact economic growth by identifying and exploiting opportunities.
- D.** 1.
 - A corporation is a legal business entity that is separate from its owners, known as shareholders.
 - Shareholders hold shares of stock in the company, which represent ownership stakes in the corporation, and shareholders may buy or sell their shares, easily transferring ownership of the business.
 - The most notable aspect of a corporation is its limited liability, meaning that shareholders are not personally liable for the corporation's debts and legal obligations.
 - This separation of personal and corporate assets provides a level of financial protection for shareholders.
2. Entrepreneurship development refers to the process of enhancing and honing the knowledge, skills, and resources of individuals to create and manage successful businesses or ventures. It involves activities and initiatives intended to foster entrepreneurial ability and support individuals in starting, growing, and managing their own businesses.
3.
 - An entrepreneur is a person who recognises potential opportunities, accepts the uncertainties of both successes and failures, and establishes and manages enterprises or initiatives to fulfil the needs of the market.
 - They take the initiative of starting and managing a business with the goal of making a profit. They often possess the qualities of innovation, vision, and a strong work ethic, seeking to bring their ideas to life, create value, and achieve financial success.
 - Entrepreneurs make a substantial impact on our society through the generation of employment, the provision of products and services, and an increase in our country's national income.
- E.** 1. There are four categories of businesses based on their activities:
- **Service business:** A service business is a type of business activity that primarily offers services that are not tangible in nature. These services are not in the form of physical goods. They cannot be touched but are experienced by the customers. In a service business, the primary focus is on delivering expertise, skills, knowledge, labour, or solutions to meet the specific needs and demands of the clients or customers.
 - **Manufacturing business:** A manufacturing business is a type of enterprise that produces goods through various processes, often involving raw materials, labour, and machinery. Manufacturing firms acquire the necessary raw materials for the production or creation of goods, which are then processed into finished products with the help of labour and machinery. These finished goods are then distributed to wholesalers, who, in turn, supply them to retailers, which in turn make them available to customers in the market.

- **Merchandising business:** A merchandising business is the type of enterprise that buys products or goods from manufacturers at a wholesale rate (price paid for buying products in bulk) and sells them to consumers or other businesses at the retail price (price paid to buy an individual entity). The primary focus of a merchandising business is on buying, stocking, and reselling physical products rather than producing or manufacturing them. The difference between the wholesale rate and the retail rate is what a merchandising unit earns as a profit.
 - **Hybrid business:** A hybrid business is an enterprise that combines elements from the three business activities, i.e., service, manufacturing, and merchandising, to create a unique and adaptable approach to achieving its goals. Hybrid businesses blend characteristics from two or more types of businesses to address specific market needs or opportunities.
2. There are specific characteristics that define successful entrepreneurship.
- **Ability to take a risk:** Any new business initiative carries a considerable chance of failure. As a result, being a successful entrepreneur requires zeal and the ability to weigh the pros and cons of taking risks. Only then will the entrepreneur be able to take the desired risk.
 - **Innovation:** The entrepreneur's idea should be an innovative and pathbreaking approach towards the opportunity identified. This will increase the chance of making a good profit.
 - **Leadership skills:** Entrepreneurs need to lead their teams and inspire others to share their vision. Effective leadership and communication skills are essential.
 - **Open-mindedness:** Open-mindedness is an essential quality for entrepreneurs as it allows them to be receptive to new ideas, feedback, and different perspectives. They see every event and situation as a business opportunity. For instance, during demonetization, there were companies that took advantage of the circumstances and increased their business. Such companies also understood the seriousness of the situation and improved the necessity for online transactions.
 - **Resilience and persistence:** Entrepreneurs bounce back from setbacks and failures. Entrepreneurship is full of ups and downs, and the ability to persevere in the face of adversity is crucial.
 - **Vision:** Successful entrepreneurs have a clear vision for their business. They can see opportunities and future possibilities.
 - **Adaptability:** Successful entrepreneurs adapt to changing circumstances. They are flexible and open to change. They can pivot when necessary and adjust their strategies in response to changing market conditions.
3. Entrepreneurship offers a range of potential rewards, both intrinsic and extrinsic, that can motivate individuals to start and grow their own businesses.
- **Financial rewards:** Successful businesses can generate profits and provide entrepreneurs with substantial income and wealth.
 - **Independence and autonomy:** Entrepreneurship allows individuals to be their own boss. They have the freedom to make decisions, set their own direction, and have control over their work.
 - **Creativity and innovation:** Entrepreneurs have the opportunity to bring their creative ideas to life and innovate in various industries. This freedom to innovate and create can be personally fulfilling.
 - **Personal growth:** Running a business often requires individuals to continuously learn, adapt, and grow. Entrepreneurship can lead to personal development, improved problem-solving skills, and increased self-confidence.
 - **Impact and fulfilment:** Many entrepreneurs are motivated by the desire to make a positive impact on their communities, industries, or society as a whole. A sense of purpose and fulfilment can be a significant reward.
 - **Recognition and reputation:** Successful entrepreneurs often gain recognition and build a strong reputation in their industries. This recognition can open doors to new opportunities and partnerships.
 - **Building of legacy:** Entrepreneurs have the potential to leave a lasting legacy, whether it's through their innovative products or services, the businesses they build, or the positive impact they have on their communities.

F. Competency-based questions.

1. Ragini should own a sole proprietorship business, as she will have complete control over all aspects of the business, from decision-making to daily operations.

She will enjoy the whole profit. She will not be obligated to follow strict legal procedures to start her boutique.

2. Rajat's role as an entrepreneur has been able to make a difference in himself and the society in the following manner:
 - **Community development:** He has been able to support the local community by generating employment opportunities for uneducated tribal women.
 - **Social change:** He has been able to make a transition towards a sustainable way of exchanging gifts and promote the idea of sustainable living.
 - **Boosts confidence:** The acceptance of his innovative ideas has helped to boost his confidence as well as the women involved in the process.
 - **Instils drive for achievement:** Having received the award and acknowledgement has motivated him to work hard and to achieve and innovate; thereby, bringing about changes and development in society.

Unit 5: Green Skills-I

Answer Key

- A.** 1. b 2. a 3. c 4. d 5. c
- B.** 1. extinction 2. sustainable 3. Natural resources 4. Green skills 5. society
- C.** 1. False. Trees are called life-savers as they convert the excess carbon dioxide in the air to oxygen.
2. True.
3. False. Blue economy focuses on promoting sustainable approaches in relation to our oceans, seas, and coastal areas.
4. True.
5. False. The green economy aids the sustainable use of natural resources and the protection of biodiversity.
- D.** 1. Environmental conservation is a practice to protect the environment and manage the natural resources for the present and future use.
Any two steps are:
- More and more trees should be planted everywhere especially in the areas where forest cover is decreasing due to human activities.
 - Saving water resources is another important aspect of environment conservation. Turn off the taps when not in use, repair leaky faucets, and install water efficient appliances.
2. A green economy is a kind of economy that aims to establish a system that balances human well-being and economic growth while reducing environmental risks and replenishing our resources. The core objective is to promote a sustainable mode of production and consumption without causing any harm to the environment.
- The United Nations Environment Program (UNEP), a global organisation that works under the United Nations, defines a green economy as “one that results in improved human well-being and social equity while significantly reducing environmental risks and ecological scarcities. It is low carbon, resource efficient, and socially inclusive”. It aims for sustainable development.
3. An ecosystem is formed when the living (plants, animals, and other organisms) and the non-living components (air, water, sunlight, soil, weather, temperature etc.) interact and co-exist with each other in a physical environment. These biotic and abiotic components are interdependent on each other. All the organisms play a specific role in an ecosystem and every ecosystem is unique in itself.
- E.** 1. Few factors which cause an imbalance in the environment are:
- **Overexploitation of resources:** With the increase in population, human beings have started to exploit natural resources at a fast pace. This is a huge threat to biodiversity and has mainly resulted in the extinction of species, further affecting, directly or indirectly, the ecological cycle of the environment.
 - **Deforestation:** Deforestation, or the thinning of forests, is one of the largest issues. This includes the removal of trees for wood products, grazing lands, and creation of huge buildings in the name of development. This is a serious environmental concern as it can result in the loss of the natural environment, damage the homes of animals, and create disturbances in the water cycle.

- **Pollution:** Pollution can harm human health and the environment in many ways. Pollutants can be found in water, air, and soil. Also, there are different kinds of pollution that impact the environment in various ways. There can be various causes of pollution like burning of fossil fuels, smoke from industries and factories, improper industrial waste disposal, increasing vehicles on roads, mining etc.
 - **Overpopulation:** Due to the rapid increase in population, resources are being over-utilised and production of wastes is also high. People make extensive use of plastic which is harmful to marine life, and street animals and also disrupts food chains, thereby, impacting the environment.
 - **Global warming:** Due to the increase in greenhouse emissions, temperatures are rising, glaciers and polar ice caps are melting, and various species are becoming extinct. The level of carbon dioxide and other harmful gases like methane, nitrous oxide etc. is increasing in the air which also leads to the depletion of the ozone layer.
2. In India, there is a growing emphasis on embracing the principles of a green economy, placing significant importance on achieving sustainable development and transitioning towards cleaner energy sources.
- **Renewable energy:** India has set ambitious targets for renewable energy deployment, and has made significant progress in recent years. The country is now one of the leading markets for solar and wind power.
 - **Energy efficiency:** India is also investing in energy efficiency measures to reduce its energy consumption. This includes programs to promote energy-efficient appliances and buildings.
 - **Sustainable transportation:** India is promoting electric vehicles and other sustainable modes of transportation. The country is also investing in public transportation and infrastructure for pedestrians and cyclists.
 - **Sustainable cities:** India is developing sustainable cities that are more energy-efficient, less polluted, and more liveable. This includes initiatives to promote green buildings, waste management, and public transportation.
 - **Sustainable agriculture:** India is promoting sustainable agriculture practices to reduce its environmental impact. This includes initiatives to reduce the use of pesticides, improve water management, and promote organic farming. Programmes like PM KUSUM Yojana and Gobardhan Yojna are supporting green farming and sustainable practices in villages.
3. • There is a strong relation between our society and the environment. They interact and influence each other in multiple ways. This kind of interaction affects the ecological balance.
- Humans depend on natural resources for fulfilling their basic needs like food, water, and shelter. With the increase in population, human beings have modified their surroundings according to their necessities and economic benefits. These modifications have impacted our natural environment significantly.
 - Industrial development and the exploitation of natural resources, like the extensive use of fossil fuels, deforestation etc. have created an imbalance in the environment. People do not use natural resources judiciously and thus it has caused a scarcity of these resources.
 - Thus, it is important to strike a balance between both components to avoid harmful consequences.

F. Competency-based questions.

1. Kanika should start by making a few transitions in her life and advocate for the same in and around her house.
- **Reduce:** She should be mindful of what she is buying and not stacking up unnecessary things. It is an effective way to save the resources. Avoid the use of plastic and any such products which have a negative impact on our environment.
 - **Reuse:** Instead of throwing items or discarding them, she should reuse them creatively. For example, glass jars can be used for storing things, old clothes come in handy when she needs to clean her home etc. She can also share her old things with those who need them.
 - **Recycle:** Cans, paper products, glasses etc. are all recyclable materials. She should separate them from non-recyclable materials while disposing them. Community recycling efforts should also be encouraged.

2. Yogesh should highlight the importance of a green economy to Vipin:

- **Protecting the environment:** The green economy aids the sustainable use of natural resources and the protection of biodiversity. This is crucial for ensuring the well-being of our planet.
- **Consideration of moral ethics:** Adopting sustainable and responsible ways of producing and consuming aligns with ethical values and responsibilities towards future generations.
- **Innovation and technological advancement:** Striving towards a green economy encourages innovation in technologies and sustainable practices, leading to economic growth and competitiveness in a fast-evolving global market.
- **Creating job opportunities:** The transition to a green economy is expected to create myriads of new jobs in varied sectors, like waste management, renewable energy, and resource management. This in turn will boost economic growth.

IT Skills

Unit 1: Introduction to IT-ITeS Industry

Answer Key

- A.** 1. a 2. d 3. a 4. c
- B.** 1. Healthcare 2. BPM 3. Global 4. Cloud Computing
- C.** 1. True.
2. False. IT in banking allows customers to access their accounts, make transactions, and manage finances conveniently.
3. True.
4. False. It is a combination of information and technology.
- D.** 1. Following are some of the benefits of IT:
• Help in finding potential customers.
• It helps provide immediate customer service.
2. BPO stands for Business Process Outsourcing. It is a business practice in which an organisation hires a third-party service provider or vendor to perform specific business tasks.
- E.** 1. Call centres are centralised offices or set up by companies to handle a large volume of telephone inquiries, requests, or customer support needs. They are staffed with customer service representatives who are trained to assist customers, answer queries, resolve issues, and provide information about products or services.
BPO stands for Business Process Outsourcing. It is a business practice in which an organisation hires a third-party service provider or vendor to perform specific business tasks. These tasks can include customer support, technical support, data entry, human resources, accounting, and other non-core functions.
2. The BPM (Business Process Management) industry in India is a significant and rapidly growing sector within the broader ITeS industry. BPM primarily involves the outsourcing of various business processes by organisations worldwide to service providers in India, where skilled professionals handle tasks such as answering phone calls, sorting papers, and performing computer work on behalf of other companies. These tasks are often non-core functions, allowing businesses to focus on their core activities while improving efficiency and reducing costs. The BPM industry in India has indeed experienced substantial growth, contributing significantly to the country's economy.
- F.** 1. The IT-BPM industries are categorised based on several key factors:
• The specific market or region the company serves
• The types of services offered by the company
• The geographical coverage and reach of the company's services
• The extent of the company's operations and the profits it generates
2. Information technology plays a vital role in shaping the landscape of home computing within the ITeS sector. With the evolution of technology and changing work dynamics, remote work has become a cornerstone of the ITeS industry. IT solutions are the backbone of this transformation, facilitating seamless telecommuting for employees.
IT infrastructure, including secure virtual private networks (VPNs) and remote desktop solutions, enables individuals to work effectively from their homes, ensuring data security with firewalls, antivirus software, and encryption protocols.
The adoption of cloud computing allows easy access to essential tools and data, fostering collaboration and data sharing. Collaboration tools, video conferencing, instant messaging, and project management software are essential for remote teamwork.

Unit 2: Data Entry and Keyboarding Skills

Answer Key

- A.** 1. a 2. b 3. d 4. c 5. a
- B.** 1. Statistics toolbar 2. Orange 3. Lesson diagram/Statistics
4. Spacebar 5. Cursor
- C.** 1. False. The lesson statistics table displays the complete statistics of the selected lesson in the course diagram.
2. True.
3. False. The toolbar allows you to customise the primary program features.
4. True.
5. True.
- D.** 1. The status bar at the bottom contains suggestions for the current lesson. It provides tips for the user to follow while typing.
2. Typing accuracy is measured as the proportion of correctly typed characters to total characters.
3. It is used to scan codes directly from the products, books, or membership cards.
- E.** 1. The navigation keys are the keys on a keyboard used to move around a screen. These may include up, left, right, and down arrows. These allow you to navigate through documents or other information.
2. **Title Bar:** The title bar is the top-most bar of the Typing Tutor window.
The top-left corner contains three buttons for moving between basic program modes:
- **Current lesson:** Used to check the current lesson
 - **Student statistics:** Used to check the statistics of the student's typing skills
 - **Lesson editor:** Used to edit the lesson settings
- The top-right corner contains the following buttons:
- **Options:** Used to change the settings such as language, background, text colour, themes, sound, font, help, update, and log out of the Rapid Typing software.
 - **Maximise, Minimise, and Close buttons:** Maximize and minimize buttons are used to control the window size and the close button is used to close the window.
3. The upper-row keys are Q, W, E, R, T, Y, U, I, O, P. It is recommended to keep your fingers on or close to the keys at all times. Each hand and finger will only serve a range of predefined keys. Regular practice will build the muscle memory necessary to increase speed and accuracy.
- The **Left hand** should focus on the following Upper-row keys:
- The pinkie finger should be on key **Q**
 - The ring finger on **W**
 - The middle finger on **E**
 - The index finger should be on **R**
 - The index finger should be on **T**
- The **right hand** should focus on the following Upper-row keys:
- The index finger should be on **Y**

- The index finger should be on **U**
- The middle finger should be on **I**
- The ring finger should be on **O**
- The pinkie finger should be on **P**

F. 1. Rapid typing offers a variety of courses based on difficulty level, including Introduction, Beginner Experienced, Advanced, and Testing.

Introduction Course – This course covers every character on your keyboard.

Beginner Course – The key placement on a keyboard can be learned with the aid of this course. After completing this course, you won't need to look at the keyboard to know which finger to press each key. Uppercase letters, special symbols, and the numeric pad are all covered throughout the course.

Experienced Course – This will assist the user in increasing typing speed by learning commonly used syllables and words.

Advanced Course – By entering actual text, the user can practice their typing skills with the help of this course.

Testing Course – With the help of an advanced typing test in this course, the user can assess their present typing speed and accuracy, and print a copy of the results.

2. The science of typing ergonomics focuses on utilising good posture and stretching practices when using a computer to prevent frequent occupational injuries. It offers logistical support for speedy and accurate typing.

Sitting Posture

- The height of the chair should be adjusted in such a manner that your feet are flat on the floor.
- The hips should be slightly higher than the knees.
- Keep your back straight.
- The back of the chair should be somewhat inclined at a 100- to 110-degree angle.
- Ensure the keyboard is close and directly in front of you.

Unit 3: Digital Documentation

Answer Key

- A.** 1. d 2. c 3. a 4. b 5. c
- B.** 1. Document area 2. Triple-click 3. Shift+Tab 4. Font size 5. Copies
- C.** 1. True.
2. False. Writer displays red wavy lines to show spelling errors and green wavy lines to show grammatical errors.
3. False. To delete a row from a table, select Table > Rows > Delete.
4. True.
5. True.
- D.** 1. The Formatting toolbar displays the commands that are used for formatting and enhancing text, such as changing the font, changing the font size, changing the font colour, adding bullets, highlighting the text, and aligning the text and numbering.
2. Select Tools > Language > Thesaurus or just press Ctrl+F7.
3. Splitting a table means dividing it into two parts, and merging means combining two tables to form a new table.
- E.** 1. a. Home b. Ctrl+Home c. End d. Ctrl+End
2. Indentation helps users understand a given document in a better way, as it organises the text and gives structure to the paragraph.
To indent a paragraph:
• Select the paragraph you want to indent.
• On the Formatting toolbar, click the Increase Indent icon.
3. To create a table by using the Table menu:
• Open a new or existing document in Writer and then move the cursor where you want to insert the table.
• Select the Table menu in the menu bar.
• Select the Insert > Table option from the menu.
• The Insert Table dialogue box appears.
• Select the number of rows and columns you want in your table, and then click OK to insert the table in your document.
- F.** 1. She can use the Special Character feature to add characters that are not present on the keyboard.
2. He wants to change the page formatting. He can get these options by selecting the Format menu and then the Page option from the menu bar.

Unit 4: Electronic Spreadsheet

Answer Key

- A.** 1. a 2. d 3. a 4. b 5. c
- B.** 1. File 2. Active 3. MIN() 4. Labels 5. Category
- C.** 1. True.
2. False. Ctrl+V is the shortcut key to paste data.
3. True.
4. True.
5. True.
- D.** 1. A spreadsheet or worksheet is a file used for storing, organising, and manipulating data in a tabular format. A spreadsheet is used to perform calculations using formulas and functions.
2. Arguments are the values that are input to functions. These values can be in the form of text, numbers, logical values (True or False), constants, range of cells, or some other functions.
3. OpenOffice Calc uses a relative cell reference by default. When we include a cell reference or a range in a formula, OpenOffice Calc always uses a relative reference. The associated cell references are used in conjunction with relative references, which often indicate a combination of a column name and a row number. The relative reference for a cell does not contain a dollar (\$) sign.
- E.** 1. The advantages of using a chart are:
- It improves the presentation and readability of the data.
 - It can be used to make certain judgments or do analysis.
 - It facilitates the concise and simple summarisation of large data.
 - It facilitates accurate data comparison.
2. Data formatting is the process of changing the appearance and layout of a cell or a range of cells to make them easier to read and more visually pleasing. Formatting data in a cell provides an additional effect on the text. Additional effect includes changing the font style, font size, automatic wrapping, bold, underline, and italic.
3. Following are the elements of a formula:
- **References:** A cell or a range of cells included in a formula for performing calculations.
 - **Mathematical operators:** Symbols of calculations such as '+', '-', '*', '/', and '^'.
 - **Constants:** The values of the data or text that do not change.
 - **Functions:** The operations that are predefined as formulas in Calc.
- F.** 1. The Sum button
2. Charts

Unit 5: Digital Presentation

Answer Key

- A.** 1. d 2. d 3. a 4. b 5. d
- B.** 1. digital presentation 2. Cut, Paste 3. With previous 4. Font size 5. Drawing
- C.** 1. True.
2. True.
3. False. There are 5 types of views available in Impress.
4. True.
5. False. Shift+Ctrl+G is the shortcut key to group objects on a slide.
- D.** 1. Duplicating a slide is useful when you want to keep a similar structure or content across multiple slides without having to recreate each one manually.
2. Shortcut keys for bold, italic, and underline are Ctrl+B, Ctrl+I, and Ctrl+U, respectively.
3. You may save time and ensure a consistent appearance throughout your presentation by making the desired changes to the slide master.
- E.** 1. To run a slide show, follow any one of the following methods:
Press F5 to start the slide show.
OR
Click on the Slide Show icon present on the Standard toolbar.
OR
Click on the Slide Show menu, and then select Slide Show.
2. To delete a row or column, follow the steps given below:
- Select the row or column you want to delete.
 - Right-click on the selected row or column.
 - Select Row → Delete or Column → Delete to delete a row or a column.
3. Custom animation is an effect that can be applied to the objects of a slide, whereas the slide transition effect is applied to the slide. Custom animation controls the appearance of the objects and slide transition; on the other hand, it also controls how one slide transitions to the next slide during the slide show.
- F.** 1. Yes, it is possible to add notes for the speaker of the presentation. She can use Notes View to do so.
2. To adjust the width of the column, follow the steps given below:
- Move the cursor to the border line between two columns.
 - Press the left mouse button and drag to adjust the column width to the desired width.

About the Book

This two-book series on IT (Subject Code 402) has been designed for students in grades 9 and 10. The series accurately aligns to the latest CBSE curriculum, imparts theoretical knowledge, helps in developing practical skills of students and enables them to prepare effectively for examinations.

The series introduces students to fundamental concepts in digital documentation, spreadsheet, presentation, database management, and internet security. It also comprises units that cultivate the employability skills and vocational proficiencies of students.

Key Features

Every chapter within the book consists of the following key elements that enhance the learning process and make exam preparations more effective.

- **Remember:** Important points to aid memory and recall.
- **Error Alert:** Common misconceptions with clear solutions.
- **Did You Know?:** Fun facts related to the topic, included to captivate students' interest.
- **Key Terms:** Important terms to ensure a firm grasp of important concepts.
- **Points to Remember:** Chapter-end point-wise summary to consolidate concepts.
- **Chapter Checkup:** Chapter-end exercises containing objective and subjective questions to enable comprehensive practice of concepts.
- **Activity Time:** Classroom and laboratory-based group and individual activities for an enhanced learning experience.
- **Practical Section:** Recommended list of practical questions for active learning.
- **Viva Questions:** Reference list of viva questions to better prepare for oral examinations.
- **Projects:** Detailed projects that promote the application of knowledge.
- **Latest Sample Paper:** Sample paper, adhering to CBSE guidelines, to ensure preparedness for theoretical exams.
- **Answers Key:** Solutions to unsolved questions to support independent practice and learning.

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Uolo partners with K-12 schools to bring technology-based learning programs. We believe pedagogy and technology must come together to deliver scalable learning experiences that generate measurable outcomes. Uolo is trusted by over 10,000 schools across India, South East Asia, and the Middle East.



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